MOHAN BABU UNIVERSITY

DATA SCIENCE LAB — SAMPLE EXPERIMENT

Experiment: Data Analysis

Student Name: _____

Student ID: _____

Date: _____

Aim

To perform basic data analysis using NumPy and Pandas.

Requirements

Python, NumPy, Pandas, and a CSV dataset.

Experiment

1. Import NumPy and Pandas.

2. Create a NumPy array containing at least 10 numeric values and calculate mean, median, minimum, maximum, and standard deviation.

3. Load a CSV file into a Pandas DataFrame.

4. Display the first five rows, shape, column names, and data types.

5. Check for missing values and duplicate rows.

6. Filter rows using a condition and sort by a selected column.

7. Group by a categorical column and calculate an aggregate such as mean or count.

8. Export the cleaned data to a new CSV file.

Expected Output

A set of summary statistics, filtered/sorted rows, grouped results, and a cleaned CSV file

Conclusion

Summarize how NumPy and Pandas helped analyze the data

Note: This is a sample practice lab experiment created for learning and website testing. It is not an official university-issued lab manual.